

Shih-Ni Prim
North Carolina State University
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Professor Emily Huang and the Search Committee
Department of Statistical Sciences
Wake Forest University

Dear Search Committee,

After completing my PhD in musicology in 2016, I was fortunate to become an Adjunct Assistant Professor at Wake Forest University (WFU) and taught MUS 101, Introduction to Western Music, in Fall 2017. The experience of designing and teaching a course at such a vibrant liberal arts college was engaging and joyful. I loved teaching on a campus where education for undergraduate students is the focus and students are viewed as colleagues. Without full-time employment opportunities in musicology, I went back to school and pursued a master's degree in statistics and then a PhD. Now nearing the end of my PhD in statistics, I am beyond excited to see the opening of tenure track positions at WFU. I am ready to again teach at a liberal arts college such as WFU.

During my PhD study in statistics, I strive to take full advantage of the wide range of faculty expertise offered by my department at NC State University and work on three projects in different directions broadly connected by the concept centering uncertainty quantification (UQ). My first project, with Dr. Brian Reich, is about spatial and causal inference; we developed a fully Bayesian, spectral method to account for unmeasured spatial confounding and provide unbiased effect estimates and conservative credible intervals for multiple exposures and multiple outcomes. Originating from a summer internship at the Lawrence Livermore National Laboratory, my second project concerns surrogate modeling for computer experiments. With Dr. Kevin Quinlan (LLNL) and Dr. Annie Booth (Virginia Tech), I developed an active learning method that utilizes Gaussian processes as surrogate models and joint predictive probabilities as acquisition functions to efficiently find sampling points to locate the optimal design for contour locations from multiple computer experiments. After two projects motivated by applications, I turned to fundamentals and am currently applying possibilistic inferential models (IM) to decision theory with Dr. Ryan Martin and Dr. Jonathan Williams. Using possibility measures, IM incorporates imprecision and provides validity guarantee. These projects provide breadth in thinking and wide opportunities for current and future collaboration. After graduation, I plan to continue these directions, as detailed in my research statement.

My teaching experience in statistics thus far includes being a teaching assistant in two statistics courses and, in Summer 2025, advising six undergraduate students from NC State and two high school students from NC School of Science and Mathematics on data science projects.

This one-on-one advising experience provided me with a sneak peek of how to mentor junior researchers. With my classroom teaching in music history and advising experience in statistics, I have formed the teaching philosophy of emphasizing critical thinking and the ability to learn over learning details. I am eager to work with undergraduate and graduate students at WFU, guiding them in learning statistics as non-majors as well as conducting statistical research as majors.

Having lived in Winston-Salem for 10 years, every visit feels like homecoming. I truly hope to have an opportunity to discuss with you what I can offer to the department. If given the opportunity to work at WFU, I plan to actively seek collaboration opportunities with other departments within WFU (e.g. the medical school), build new collaborations with fellow statisticians, maintain collaborations with external institutions such as LLNL, and guide students through teaching and advising.

Sincerely,

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